

Mohammed Arshaq

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LinkedIn | GitHub | LeetCode | CodeChef

EDUCATION

BMS College of Engineering

Bachelor of Engineering in Computer Science (Data Science) **CGPA: 8.87**

Bengaluru, India

Aug 2023 – May 2027

Shaheen Falcon PU College

PUC (Class XII) — Aggregate: 89.5%

Bengaluru, India

Aug 2021 – May 2023

EXPERIENCE

Research Intern, Samsung PRISM

Jan. 2026 – Present

Samsung R&D Institute India

Bengaluru, India

- Built a unified multilingual **Grapheme-to-Phoneme (G2P) Transformer** (seq2seq) for **Hindi, Gujarati, and Marathi** on a 54.7K-entry **IndicTTS** dataset, replacing the need for 3 separate per-language models
- Designed **Acoustic Phonetic Folding**: extracted phoneme co-occurrence statistics (**PPMI**), projected embeddings via **UMAP**, and applied **K-Means clustering** guided by linguistic phonetic features to merge acoustically similar phonemes
- Reduced output phoneme vocabulary by **31.6% (57 → 39 clusters)** with negligible impact on accuracy (**PER 0.08%**, **WER 0.35%**) and speech quality (**MOS 4.77 vs. 4.78**, $p = 0.53$), validated via **Mann-Whitney U** testing

PROJECTS

GradeWise AI – Sustainable AI Assessment Platform | FastAPI, React, MongoDB, HuggingFace



- Built an AI-powered grading system for handwritten answer sheets via a **three-stage pipeline**: **Mistral OCR** → text structuring → **Qwen2.5-0.5B SLM** evaluation, with **zero cloud dependency**
- Built an async job system using **FastAPI**, **Redis** (caching + queue), and **MongoDB** (Motor) for non-blocking batch processing with real-time progress tracking
- Developed a **React + TypeScript** frontend for grading review and analytics.

TubeMind – YouTube Intelligence Platform | Next.js, FastAPI, LangChain, ChromaDB



- Built a full-stack AI platform to **chat with any YouTube video** using a **RAG pipeline**: transcripts chunked, embedded via **Gemini text-embedding-004**, and stored in **ChromaDB** for semantic retrieval
- Engineered a **LangChain** retrieval chain with timestamp-aware chunking, delivering answers with precise video **timestamp citations**
- Designed a **Next.js 15 + TypeScript** frontend with real-time chat, **video summarization**, and topic extraction; deployed on Vercel + Render

NovaChat – Real-Time Chat Application | React, Node.js, Socket.IO, Redis, Express



- Built a full-stack real-time chat app with **global and 1-on-1 messaging** via persistent **WebSocket** connections using **Socket.IO**, **React 18**, and **Node.js**
- Engineered a **Redis** native data layer using **Hashes** for sessions, **Sets** for presence tracking, and **Streams** for chat history persistence
- Implemented **session-based auth** with **bcrypt** hashing; designed a glassmorphism UI with **typing indicators** and **Cloudinary**-hosted avatars

TECHNICAL SKILLS

Languages: C, Java, Python, JavaScript, TypeScript

Web Development: React.js, Next.js, Node.js, FastAPI, Socket.IO, Tailwind CSS

AI/ML: TensorFlow, PyTorch, LangChain, HuggingFace, RAG, ChromaDB

Databases & Tools: MongoDB, PostgreSQL, Redis, Docker, Git

ACHIEVEMENTS & CERTIFICATIONS

Achievements: 1600+ LeetCode rating (**Top 20%** globally), **2-Star** CodeChef, **1500+** DSA problems solved across LeetCode, GFG, and CodeChef.

Nutanix Certified Associate (NCA) – Cloud Computing Certification

Simplilearn ML using Python – Machine Learning Certification

